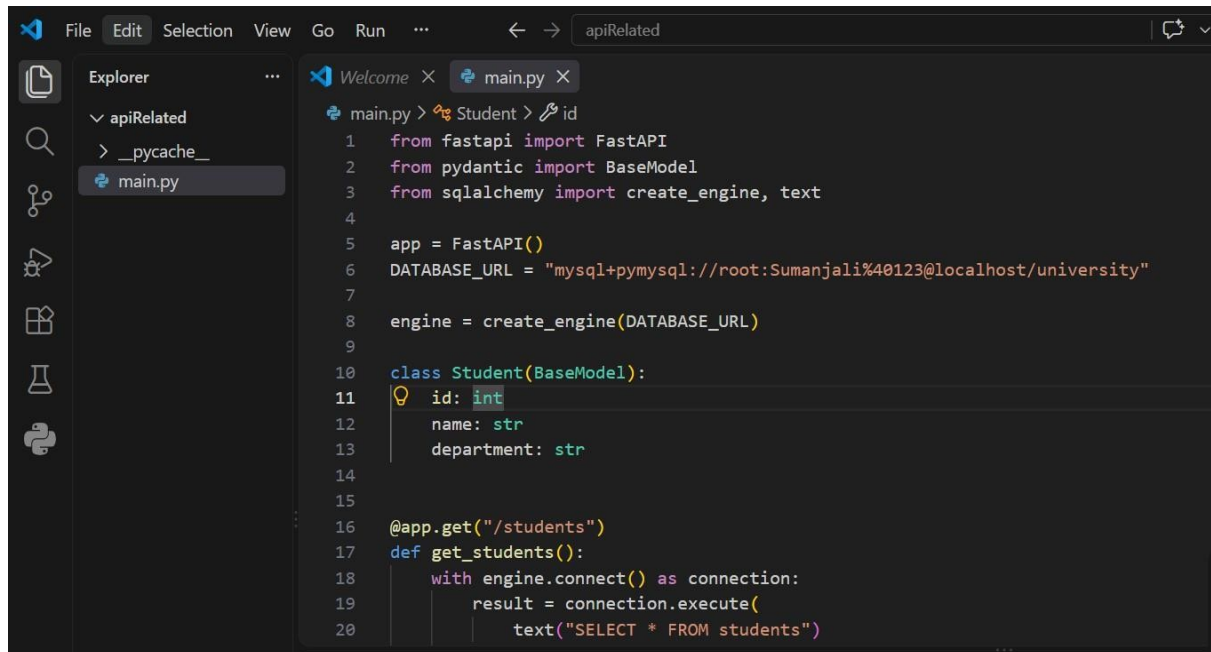
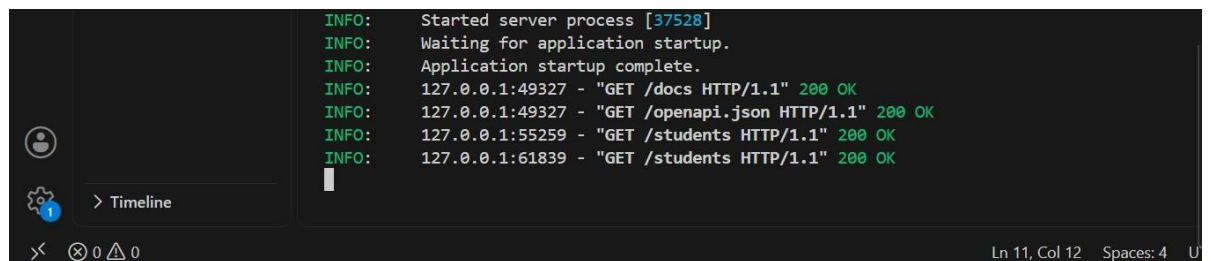


SKILL-5 K. Samanth



```
1 from fastapi import FastAPI
2 from pydantic import BaseModel
3 from sqlalchemy import create_engine, text
4
5 app = FastAPI()
6 DATABASE_URL = "mysql+pymysql://root:Sumanjali%40123@localhost/university"
7
8 engine = create_engine(DATABASE_URL)
9
10 class Student(BaseModel):
11     id: int
12     name: str
13     department: str
14
15
16 @app.get("/students")
17 def get_students():
18     with engine.connect() as connection:
19         result = connection.execute(
20             text("SELECT * FROM students")
21         )
```



```
INFO: Started server process [37528]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:49327 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:49327 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:55259 - "GET /students HTTP/1.1" 200 OK
INFO: 127.0.0.1:61839 - "GET /students HTTP/1.1" 200 OK
```



```
4
5
6 @app.get("/students")
7 def get_students():
8     with engine.connect() as connection:
9         result = connection.execute(
10             text("SELECT * FROM students")
11         )
12         students = result.mappings().all()
13         return students
14
```

default

GET /students Get Students

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code	Details
200	Response body [

Server response

Code	Details
200	Response body [{ "id": 101, "name": "Ravi", "department": "CSE" }] Response headers content-length: 45 content-type: application/json date: Wed, 09 Sep 2026 10:16:58 GMT server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

Limit to 1000 rows

1 • CREATE DATABASE university;

2 • USE university;

3 • CREATE TABLE students (

4 id INT PRIMARY KEY,

5 name VARCHAR(100),

6 department VARCHAR(100)

7);

8 • SELECT * FROM students;

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week_2_practical SQL File 2"

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid

id	name	department
101	Ravi	CSE
NULL	NULL	NULL

students 3 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	15:46:04	CREATE TABLE students (id INT PRIMARY KEY, ...	Error Code: 1050. Table 'students' already exists	0.016 sec
7	15:46:11	INSERT INTO students (id, name, department) VALUE...	1 row(s) affected	0.015 sec
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQL File 2"

```

4      id INT PRIMARY KEY,
5      name VARCHAR(100),
6      department VARCHAR(100)
7  );
8  • SELECT * FROM students;
9  • INSERT INTO students (id, name, department)
10 VALUES (101, 'Ravi', 'CSE');
11 • SELECT * FROM students;

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

24
25
26 @app.post("/students")
27 def create_student(student: Student):
28     with engine.begin() as connection:
29         connection.execute(
30             text("""
31                 INSERT INTO students (id, name, department)
32                 VALUES (:id, :name, :department)
33             """),
34             {
35                 "id": student.id,
36                 "name": student.name,
37                 "department": student.department
38             }
39         )
40
41     return {
42         "message": "Student created",
43         "student": student
44     }

```

POST /students Create Student

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value Schema

```
{
  "id": 102,
  "name": "Priya",
  "department": "ECE"
}
```

Execute

Clear

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "id": 102,
    "name": "Priya",
    "department": "ECE"
  }'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code	Details
200	Response body <pre>{ "message": "Student created", "student": { "id": 102, "name": "Priya", "department": "ECE" } }</pre> Download Response headers <pre>content-length: 84 content-type: application/json date: Thu, 10 Sep 2026 03:59:17 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week_2_practical Skill_week-5

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information: No object selected

Result Grid

id	name	department
101	Ravi	CSE
102	Priya	ECE
NONE	NONE	NONE

students 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	15:46:11	INSERT INTO students (id, name, department) VALUE...	1 row(s) affected	0.015 sec
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	09:31:13	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.032 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

44     }
45     @app.put("/students/{student_id}")
46     def update_student(student_id: int, student: Student):
47         with engine.begin() as connection:
48             connection.execute(
49                 text("""
50                     UPDATE students
51                     SET name = :name,
52                         department = :department
53                     WHERE id = :student_id
54                     """),
55                 {
56                     "student_id": student_id,
57                     "name": student.name,
58                     "department": student.department
59                 }
60             )
61
62         return {
63             "message": "Student updated",
64             "student_id": student_id,
65             "student": student
66         }

```

default

GET

/students

Get Students

POST

/students

Create Student

PUT

/students/{student_id}

Update Student

Parameters

Cancel

Reset

Name	Description
student_id * required	
integer	101
(path)	

Request body * required

application/json

Edit Value | Schema

```
{
  "id": 101,
  "name": "Ravi Kumar",
  "department": "AIML"
}
```

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/101' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 101,
    "name": "Ravi Kumar",
    "department": "AIML"
  }'
```

Request URL

http://127.0.0.1:8000/students/101

Server response

Code	Details
200	Response body <pre>{ "message": "Student updated", "student_id": 101, "student": { "id": 101, "name": "Ravi Kumar", "department": "AIML" } }</pre> Download Response headers <pre>content-length: 107 content-type: application/json date: Thu, 10 Sep 2026 04:05:07 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week_2_practical Skill_week-5

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information: No object selected

Result Grid

id	name	department
101	Ravi Kumar	AIML
102	Priya	ECE
NULL	NULL	NULL

students 5 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	09:31:13	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.032 sec / 0.000 sec
10	09:35:55	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
64     "student_id": student_id,
65     "student": student
66 }
67 @app.delete("/students/{student_id}")
68 def delete_student(student_id: int):
69     with engine.begin() as connection:
70         connection.execute(
71             text("""
72                 DELETE FROM students
73                 WHERE id = :student_id
74             """),
75             {
76                 "student_id": student_id
77             }
78         )
79     return {
80         "message": "Student deleted",
81         "student_id": student_id
82     }
83
```

Ln 83, Col 6 Spaces: 4 UTF-8 CRLF Python Python 3.14.7 Go Live

DELETE

/students/{student_id}

Delete Student

Parameters

Cancel

Name	Description
student_id * required	
integer	101
(path)	

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n'http://127.0.0.1:8000/students/101' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/students/101

Code

Details

200

Response body

{\n "message": "Student deleted",\n "student_id": 101\n}

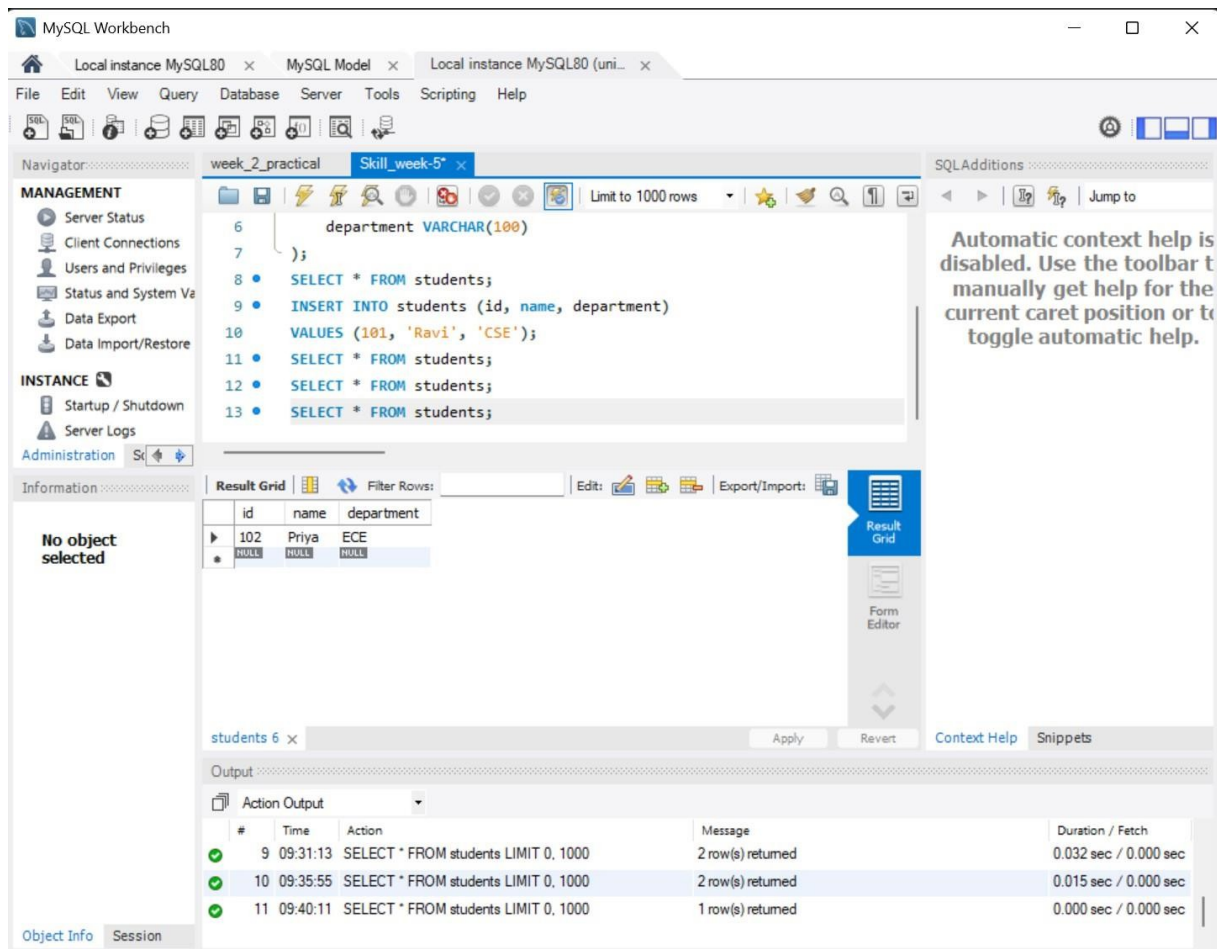
Download

Response headers

content-length: 46\ncontent-type: application/json\ndate: Thu, 10 Sep 2026 04:09:47 GMT\nserver: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links
	Media type	
	application/json	
	Controls Accept header.	
	Example Value Schema	
	"string"	
422	Validation Error	No links
	Media type	
	application/json	
	Example Value Schema	



FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET	/students	Get Students	▼
POST	/students	Create Student	▼
PUT	/students/{student_id}	Update Student	▼
DELETE	/students/{student_id}	Delete Student	▼

Schemas

HTTPValidationError > Expand all object

Student > Expand all object